

Experimental Performance Evaluation of Wireless 802.11b Networks

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Abstract

According to the IEEE 802.11b/g standard (WiFi) should have a range of 300-500 ft (about 152m). The 2.4GHz frequency band used by WiFi devices is divided into 11 (in FCC) (or 13 in ETSI) partially overlapping channels. It is claimed that by using frequency bands that are at least four channels apart (non-overlapping channels) interference between those channels is avoided.

We carried out two types of experiments to measure the effect of the distance and the claimed non-overlapping channels on the bit rate. From the first study, we experimentally derived a model of distance versus throughput. The second study has confirmed that the interference between the channels cannot be avoided even when using non-overlapping channels. The experiments also showed that by increasing the distance between the networks that are using non-overlapping channels, the effect of the interference decreases.

Key words: 802.11, Performance evaluation, Log-distance model, Non-overlapping channels.

1. Introduction

Wireless networks are usually easy to install and to maintain. They are cheaper than wired networks and easier to connect to. Wireless communication is available through various reasonably priced devices. This makes it easy for anyone to find a network and connect to it to browse the internet or to check his/her email in airports, coffee shops and even on the street. The driving forces for developing wireless devices are the economy, the military, and people's natural need to communicate with each other.

There are different types of these widely spread technologies. The most common ones are Wi-Fi, Bluetooth, and the emerging WiMAX. These

technologies have different attributes such as their range, bit rate, price, penetration, and their resistance to interference. Those attributes are affected by various environmental conditions.

The actual performance of wireless networks takes into account the different surrounding conditions that are not mentioned with the specification of the commercial products for marketing purposes. Therefore, our objective in this paper is to study experimentally the effect of different circumstances on Wi-Fi wireless networks. We establish experimentally a model to capture the effect of the distance on the bit rate. We show also the effect of the claimed non-overlapping channels on the throughput even when the networks are few meters far from each other's.

The rest of the paper is organized as follows: After the introduction, we provided some background information. We then discussed previous related work. After that, we presented our experimental studies then we finally concluded and gave some recommendations.

2. Background

The main Wi-Fi technologies, IEEE 802.11 are currently a radio-based wireless LAN. The 802.11 has 3 main types 802.11a, 802.11b, 802.11g, and the evolving 802.11n [1-2]. 802.11b uses Direct Spread Spectrum Sequence (DSSS) Modulation scheme while 802.11g uses DSSS and Orthogonal Frequency-Division Multiplexing (OFDM). Some main information about those standards is mentioned in table 1.

In our study, we targeted 802.11b standards, which is the most commonly used. These standards use the 2.4-2.485 GHz radio frequency range, which is divided into 11 partially overlapping channels, see Figure 1. A WLAN can use a maximum of three non-overlapping channels; these are channels 1, 6, and 11 as argued in [3].

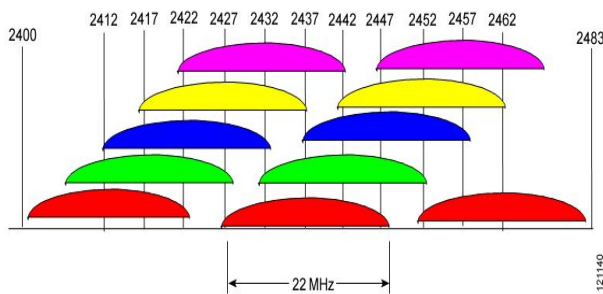


Figure 1. IEEE 802.11 channels

This standard is easier to setup, and has a wider range, between 300ft and 500ft (about 91m-152m) as it operates in the 2.4-2.485 GHz range. However, because of this range of frequency, 802.11b can interfere with different devices like home appliances such as microwaves and cordless phones.

802.11b has a speed of 1-11 Mbps, which is slower than 802.11a standard's speed. This standard has 11 channels but can broadcast on one channel only at a time.

802.11b is much cheaper than the 802.11a and is the most common technology used nowadays. IEEE 802.11 [4] employs Distributed Coordination Function (DCF) as primary mechanism to access the medium. DCF is a random access protocol based on Carrier Sensing Multiple Access with Collision Avoidance (CSMA/CA).

Table 1 Wi-Fi common technologies

	802.11a	802.11b	802.11g
Modulation Schemes	OFDM	DSSS	DSSS & OFDM
frequency Operation	5.15-5.135 GHz 5.725-5.825 GHz	2.4-2.4835 GHz	2.4-2.4835 GHz
Bit rate	1-54 Mbps	1-11 Mbps	1-54 Mbps
Range	50-150ft	300-500ft	300-500 ft

3. Related Work

Few works have been done to experimentally study the performance of IEEE 802.11 wireless networks. In [5] a characterization of the indoor radio propagation has been presented for fading, multipath, and background interference using 2.4 GHz band. However, the study did not consider the distance as a parameter. In [6] a methodology for

characterizing different vendors 802.11 network cards. In addition, here again the distance parameter is not taken into account. In [7] the authors study the signal attenuation in IEEE 802.11 wireless LANs. They conducted experiments to validate the *log-distance* model proposed in [8]. *Log-distance model is used to predict signal attenuation versus distance.* In this paper, we extend this work to predict bit rate versus distance. From the application point of view, bit rate is essential. Another study of our interest is to confirm the interference of non-overlapping channels. It has been widely accepted that multiple IEEE 802.11b/g can transmit without interference if they use non-overlapping channels (channel that are at least 25 MHz apart from each others), as an example channels 1, 6, 11. In practice, it has been shown in [9] that for close enough (few centimeters distance) wireless networks even if they use non-overlapping channel cross interference will be present. The study has been conducted for mesh multi-hop networks [9-10]. In our experimental study, we extend the measurement for networks using non-overlapping channel that are few meters distance and we confirm the interferences of the non-overlapping channels.

4. Experimental Studies on IEEE802.11 b

In what follows, we conduct two different studies on the IEEE802.11 WLAN. The first study focuses on the throughput (bit rate) versus the distance in Ad-Hoc networks. The second study consists of measuring the bit rate as an effect of the distance on networks using non-overlapping channels.

4.1 Throughput versus Distance

To conduct this experiment we have used widely used equipments.

Hardware:

- Toshiba satellite L10 with Intel(R) PRO/Wireless 2200BG Network Connection WNIC.
- Toshiba Tecra A4 with Intel(R) PRO/Wireless 2200BG Network Connection WNIC.

Software:

Iperf [12] is free client/server software used to generate traffic within a network. There are many options available to control the generated traffic.

Setup and Results:

The experiment was conducted at an open and empty parking lot, see Figure 2, on a weekend to avoid any interference and eliminate barrier effect.



Figure 2. The experiment at the parking lot

We use Ad-Hoc connection between the two previously mentioned laptops. *Iperf* software was running as a server on one laptop and as a client on the other. We started with a distance of 0m and used *Iperf* to generate packets. The experiment was repeated every 15m.

From Figure 3, we can notice that increasing the distance between the server node and the client caused a decrease in the throughput. When the nodes were separated by 0m the throughput was about 17 Mbps. The throughput decreased by more than 50% when the nodes are separated by 60m. At 100m the throughput was less than 5Mbps. Inspired by the log-distance model, we used regression techniques and the data shows a strong correlation indicated by the correlation coefficient ($R^2 = 0.940$), with the logarithmic function:

$$\text{Throughput}(d) = -6.5618 \ln(d) + 17.963 \quad (1)$$

Where the throughput is expressed in bit per seconds and the distance d in meters.

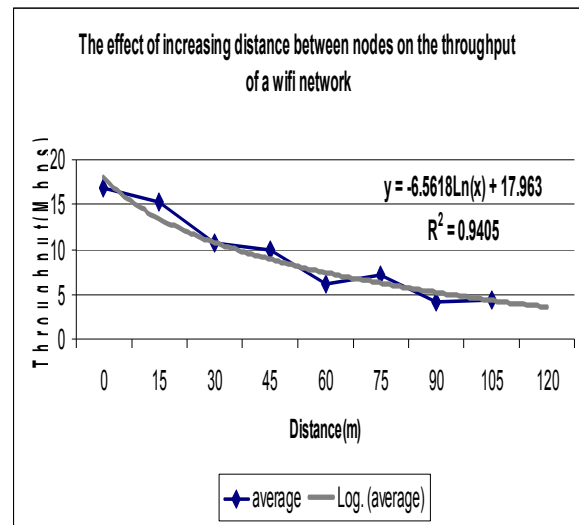


Figure 3. The throughput versus distance

4.2 Distance on Non-Overlapping Channels

4.2.1 Experiment 1

In this experiment, we concentrated on the effect of non-overlapping channels

Hardware used:

- Toshiba satellite L10 with Intel(R) PRO/Wireless 2200BG Network Connection WNIC.
- Toshiba Tecra A4 with Intel(R) PRO/Wireless 2200BG Network Connection WNIC.
- 2 Fujitsu Siemens ESPRIMO desktop with U.S. Robotics WNIC.

Setup and Results:

We created two Ad-Hoc networks; *Network1* and *Network2*. The networks used non-overlapping channels (1 and 6, 6 and 11, 1 and 11). We measured their throughput when each network sent data on its own, and when both of the networks sent simultaneously. The two networks are placed near each other (less than 1 meter apart).

Figure 4, shows the average throughputs obtained by the experiment.

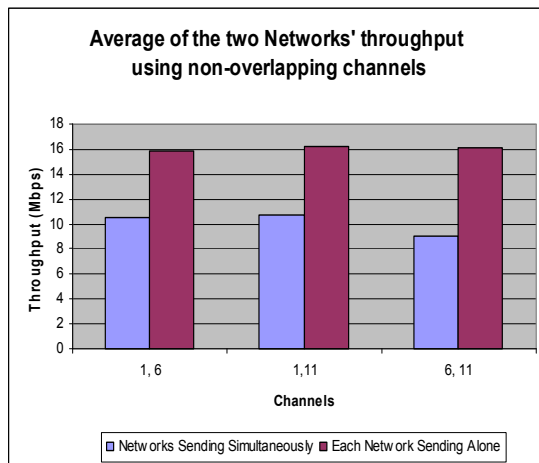


Figure 4. Two networks Net1 and Net2

It is claimed that channels 1, 6 and 11 are non-overlapping channel but in practice, we have confirmed that it is not true, they do interfere. Figure 4 shows the interference between non-overlapping channels, which affects the networks' throughput. It shows that the interferences reduce the throughput to almost 50%.

4.2.2 Experiment 2

Now let us turn our attention to mobility in the same IP subnet. In order to increase the physical range of a wireless LAN, companies and universities often deploy multiple Access Points within the same IP subnet by interconnecting the APs using a hub and giving them the same Service Set Identifier (SSID) [1]. In this experiment, we tried to find a compromise that guaranties low interference and good network coverage.

Hardware used:

- Two Linksys Wireless-G Access point (WAP54G).
- A Toshiba Satellite L10 with Intel(R) PRO/Wireless 2200BG Network Connection WNIC.
- Toshiba Tecra A4 with Intel(R) PRO/Wireless 2200BG Network Connection WNIC.
- Toshiba Satellite A100-785 with Intel 11/a/b/g Network Connection WNIC.

- Dell Inspiron 6400 Notebook with Intel(R) PRO/Wireless 3945 ABG Network Connection WNIC.

Setup and Results:

Two infrastructure wireless networks were created, Network1 and Network2. Each one contains two laptops, a client and a server, and an AP. One AP used channel 1 and the other used channel 6, which are non-overlapping channels. Both APs were connected to a Local Area Network and the laptops ran Iperf software.

The two APs are placed at different distances from each other. The throughput (bit rate) of the networks is measured. We started from a distance of 20m and continued moving the APs apart from each other.

From the previous experiment, we find that non-overlapping channels interfere. Therefore, this experiment was designed to find out if an appropriate distance between the two APs could be obtained such that the network coverage is continuous and the interference is reduced.

Table 2, shows that at the starting point of the experiment, when the APs were 20m apart from each other, the two networks were interfering to some extent. As we went on moving the APs, we found out that, in our environment, around 25m was a good distance to keep between the two APs to have minimum overlapping and interfering between them. In fact, the higher the distance separating the APs the lower the interference.

Table 2. Throughput of networks varying the distance

Distance(m)	20	21	23	25
Network1 Throughput(Mbps)	9.23	8.26	10.5	10.7
Network2 Throughput(Mbps)	9.33	11.6	11.7	11.6
Average Throughput	9.28	9.93	11.1	11.15

5. Conclusion

We have conducted different experimental studies and derived a log-distance model to predict bit rate versus distance for wireless LAN. This model could be used to predict the throughput versus distance in the absence of other interference.

We have also confirmed the interference of WiFi non-overlapping channels even if the networks are several meters apart. By combining our findings in the experiments on non-overlapping channels, we concluded that using non-overlapping channels would not eliminate the interference between networks available in the same area. However, a good compromise can be obtained to minimize the interference while maintaining continuous network connectivity. The optimal distance to keep between APs can be found by doing experiments on the target area or environment.

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